From Butterfly Came the Kraken: Structural Indicators of Digital Settlement, Liquidity Compression, and Market-Structure Transition in U.S. Equity Markets

Abstract

This paper evaluates whether a cluster of issuer filings, capital-structure modifications, clearing-agency rule changes, and alternative trading infrastructure developments observed during 2025 collectively indicate a directional transition toward digitally verified securities settlement in U.S. equity markets. The study integrates market microstructure theory, regulatory analysis, and a replicable empirical framework: the Settlement Acceleration Index (SAI). The framework is designed to convert qualitative settlement-modernization signals into a measurable, backtestable indicator. The paper further develops a Monte Carlo squeeze-risk model incorporating borrow scarcity, dealer hedging, and settlement compression.

The paper distinguishes carefully between **confirmed facts**, **interpretive analysis**, and **forward-looking inference**. Its core conclusion is not that coordinated intent has been proven, but that multiple observable developments appear directionally consistent with accelerated, more auditable settlement architecture.

1 Executive Summary

This report reviews multiple corporate filings, financing activities, and clearing-agency rule changes occurring between the first and fourth quarters of 2025. The central question is

whether these events collectively align with an ongoing transition toward digitally verified equity settlement in U.S. markets.

Confirmed observations include the existence of a sequence of private capital formations through Dominari IM LLC Series funds, shelf-registration activity and strategic announcements from Dominari Holdings Inc., capital-structure adjustments by certain issuers including warrant-related actions, and regulatory filings from clearing agencies including NSCC and FICC.

Interpretive analysis suggests that these developments may have implications for collateral efficiency, delivery sensitivity, and the transparency of beneficial ownership.

Inferred implications include the possibility that U.S. equity markets are entering a structural transition from traditional delayed settlement toward more transparent, digitally auditable ownership systems. That transition, if realized, could reduce settlement latency, improve ownership verification, and alter liquidity dynamics for heavily shorted or structurally complex equities.

2 Findings Classification Framework

This paper uses the following evidentiary categories throughout:

2.1 Confirmed

A claim is classified as **Confirmed** when it is directly supported by an identifiable public filing, rule release, memorandum, or disclosed market document. Examples include the existence of a filing, a rule proposal, a public memorandum, or a dated issuer action.

2.2 Interpretive

A claim is classified as **Interpretive** when it derives from reasoned analysis of confirmed facts, but the conclusion itself is not explicitly stated in the underlying source. Examples

include assessments of likely market-structure relevance, delivery sensitivity, or balance-sheet consequences.

2.3 Inferred

A claim is classified as **Inferred** when it extends beyond directly observable facts into scenario analysis, directional hypothesis, or forward-looking structural implications. Such claims are presented as plausible analytical conclusions rather than established fact.

3 Introduction

Modern securities markets remain shaped by the timing gap between trade execution and final settlement. Even where front-end execution appears instantaneous, beneficial ownership confirmation, collateral substitution, securities lending, and netting mechanics continue to operate with frictions that preserve opportunities for balance-sheet intermediation and synthetic exposure formation.

Confirmed: settlement design materially affects the timing of final ownership recognition and collateral processing in securities markets.

Interpretive: longer settlement intervals may provide more room for balance-sheet intermediation, delayed reconciliation, and substitute delivery mechanisms.

Inferred: as market infrastructure evolves toward near-real-time settlement and digitally auditable ownership, those degrees of freedom may compress.

This paper examines whether a set of events observed between Q1 and Q4 2025 can be interpreted not as isolated developments, but as components of a common structural direction. These events include: (i) clustered private-capital formations across a thematic infrastructure stack, (ii) shelf and warrant actions by issuers, (iii) options deliverable changes, (iv) DTCC-related rule modernization, and (v) the maturation of tokenized alternative trading frameworks.

The paper does not claim that these events prove centralized coordination. Instead, it evaluates whether they reveal measurable *structural alignment* with accelerated, digitally verifiable settlement architecture.

4 Background and Scope of Review

The objective of this study is to determine whether recent corporate and regulatory actions form a coherent directional trend toward digital or tokenized market infrastructure.

4.1 Research Questions

The analysis focuses on four primary questions:

1. Do issuer filings and financing actions during 2025 exhibit a common directional relationship with digital settlement readiness?
2. Do clearing-agency and ATS developments indicate operational preparation for more auditable or accelerated ownership verification?
3. Can those developments be transformed into a measurable composite indicator?
4. How would faster settlement and more rigid delivery verification interact with short exposure, borrow scarcity, and dealer hedging dynamics?

4.2 Scope

The scope is limited to publicly observable filings, rule changes, disclosed platform materials, and a generalized market-structure modeling approach. The paper develops a research architecture rather than claiming definitive proof of implementation or intent.

5 Selected Literature and Regulatory Context

A substantial literature links settlement design, collateral mobility, and market resilience. The BIS and CPMI have noted that distributed ledger technology may improve post-trade processing, clearing, and settlement efficiency, while also changing the trade-offs around liquidity, interoperability, and legal finality (BIS and CPMI, 2017, 2023; Bech et al., 2020). The SEC's shortening of the U.S. standard settlement cycle to T+1 reflects a related policy objective: reducing credit, market, and liquidity risk associated with open transactions (SEC, 2023). IOSCO has separately emphasized that tokenized financial assets and broader crypto-asset market structures raise issues of market integrity, investor protection, custody, governance, and operational resilience (IOSCO, 2023, 2025). IMF work on tokenization and financial market structure likewise highlights that tokenization may reduce certain frictions and process costs while preserving important questions around competition, intermediation, and stability transmission (IMF, 2025, 2024).

In parallel, clearing and options infrastructure continued to evolve in 2025. DTCC materials for SR-NSCC-2025-013 describe enhancements to the CNS Fails Charge methodology, while SR-FICC-2025-021 describes a new ACS Triparty Service in FICC's Government Securities Division (NSCC, 2025; FICC, 2025). OCC information memoranda from October 2025 document adjusted GME option deliverables and delayed settlement mechanics tied to the GME1 class (OCC, 2025). SEC materials also indicate ongoing engagement with tZERO-related digital asset securities custody and ATS infrastructure (SEC, 2025).

Confirmed: official literature and regulatory materials support the proposition that settlement speed, collateral design, and digital ownership systems are active policy and market-structure topics.

Interpretive: these references provide a conceptual basis for analyzing 2025 issuer and infrastructure events through a settlement-modernization lens.

6 Data Sources

The framework is designed to ingest data from the following source categories:

a. SEC EDGAR filings, including Form S-3, 10-Q, 8-K, and Form D.

b. DTCC, NSCC, and FICC rule filings and related notices.

c. OCC memoranda relating to deliverable adjustments and corporate actions.

d. CFTC positioning reports and weekly swaps disclosures where relevant.

e. Securities lending and borrow-availability datasets.

f. Public ATS disclosure materials relevant to tokenized or digitally registered securities.

g. Market data inputs: price, volume, realized volatility, implied volatility, option open interest, and short-interest proxies where available.

7 Conceptual Framework

The underlying thesis is that as settlement systems move toward more auditable and compressed post-trade workflows, the ability to maintain opaque synthetic exposure may decline.

Confirmed: faster settlement and tighter verification alter operational post-trade mechanics.

Interpretive: those altered mechanics are likely to affect how leverage, securities lending, and delivery-sensitive positions are maintained.

Inferred: the decline should be most visible in securities where delivery obligations are unusually sensitive to capital-structure changes, ownership scarcity, or embedded derivative claims.

To formalize this, the paper introduces the **Settlement Acceleration Index (SAI)**.

7.1 Definition of SAI

Let

$$SAI_t = \alpha R_t + \beta D_t + \gamma C_t + \delta T_t + \varepsilon I_t \tag{1}$$

where:

- R_t = regulatory modernization score at time t
- D_t = digital trading / ATS readiness score
- C_t = clearing and collateral compression score
- T_t = tokenized or digitally mapped issuance activity score
- I_t = issuer-specific capital-structure sensitivity score
- $\alpha, \beta, \gamma, \delta, \varepsilon$ are researcher-selected or optimized weights

A higher value of SAI_t indicates stronger structural alignment with digitally auditable or accelerated settlement.

7.2 Interpretation

The SAI is not a forecast of price. It is a structural indicator measuring whether the market environment is becoming less compatible with opaque balance-sheet intermediation and more compatible with direct ownership verification. In practice, the SAI may be used as:

a. a policy-monitoring tool,

b. an event-study regressor,

c. a screening variable for securities vulnerable to settlement compression,

d. an input into squeeze-risk simulations.

8 Structural Stack Analysis of Dominari IM LLC Series Vehicles

Between April and June 2025, Dominari IM LLC filed a sequence of Form D notices covering multiple series vehicles tied to frontier technology exposures.

Confirmed: the filings exist and the vehicles were associated with identifiable companies and sectors.

Interpretive: although each series is legally distinct, the combined exposure set resembles a vertically layered infrastructure stack.

Table 1: Illustrative Infrastructure Stack from Series Vehicle Exposures

Series	Core Entity	Sector	Functional Interpretation
V, IX, XV, XVII	SpaceX Holdings	Communications	Data / network backbone
VI, XVI	xAI LLC	AI systems	Decision / ranking layer
VII, XI	Cerebras Systems	AI hardware	Training compute layer
XII	Groq Inc.	Inference silicon	Real-time compute layer
XIV, XIX	Anduril Industries	Autonomy / security	Sensing / defense layer
XVIII	Zipline Intl.	Logistics	Last-mile distribution layer
VIII	EVTEC Automotive Ltd.	Mobility	Transport hardware layer
III	Aquyre Biosciences Inc.	Diagnostics	Healthcare vertical
I	Cordio Medical Ltd.	Med-tech sensors	Remote monitoring layer
X	Diasome Pharma Inc.	Therapeutics	Health data integration
XX	ABTC Corp.	Battery materials	Energy substrate

Inferred: clustered capital formation in communications, intelligence, logistics, and energy may be directionally consistent with platforms that could support digital commerce, identity-linked ownership, or data-verified retail ecosystems. That implication remains analytical rather than directly disclosed.

9 Chronology of Key Events

Table 2: Indicative Timeline of Structural Market Events

Date / Period	Event	Structural Relevance
Apr–Jun 2025	Dominari IM LLC Series Form D filings	Capital formation across infrastructure sta
Jul 2025	tZERO disclosure materials	Tokenized ATS readiness
Aug 11 2025	Dominari Holdings 10-Q	Operating scope confirmation
Aug 28 2025	Dominari Holdings S-3 shelf	Capital flexibility authorization
Aug 29 2025	Crypto Advisory Board announcement	Digital asset orientation
Q3 2025	NSCC/FICC modernization filings	Post-trade system compression
Oct 7 2025	Issuer warrant action under active shelf	Delivery-sensitive capital structure
Oct 7 2025	OCC deliverable adjustment memorandum	Options settlement consequences

Confirmed: the chronology contains dated, public events.

Interpretive: the sequence is analytically useful because it allows event-study modeling and component scoring within the SAI framework.

Inferred: temporal clustering may indicate structural alignment, but does not independently prove coordination or joint planning.

10 Capital-Structure Transmission Channel

Capital-structure changes matter because settlement architecture becomes more binding when a security embeds additional delivery-sensitive claims, such as warrants, conversion rights, or complex deliverable packages.

Confirmed: corporate actions can alter the contractual deliverables associated with listed derivatives.

Interpretive: if a security's options deliverable changes from a plain common-share

claim to a mixed package involving shares and warrants, then market makers, short sellers, and clearing systems face higher reconciliation sensitivity.

Inferred: under conditions of tight borrow, these changes may alter hedging demand and settlement complexity simultaneously.

10.1 Basic Dealer Delta Exposure

Dealer delta exposure can be approximated as:

$$\Delta_t^{dealer} = \sum_{i=1}^{N} n_{i,t} \cdot \Delta_{i,t} \cdot m_i \tag{2}$$

where:

- $n_{i,t}$ = number of contracts in option series i at time t
- $\Delta_{i,t}$ = option delta
- m_i = contract multiplier / deliverable scaling factor

When corporate actions modify deliverables, m_i changes mechanically, affecting dealer hedge sizing.

10.2 Gamma and Reflexivity

Gamma is defined as:

$$\Gamma_{i,t} = \frac{\partial^2 V_{i,t}}{\partial S_t^2} \tag{3}$$

Confirmed: gamma mathematically measures convexity in option value relative to spot price.

Interpretive: aggregate dealer gamma helps determine whether hedging dampens or amplifies price moves.

Inferred: in negative-gamma regimes, upward price movement may force additional buying, especially when borrow conditions are tightening.

11 Settlement Compression and Synthetic Leverage

Settlement acceleration reduces the time window in which market participants can rely on delayed reconciliation or collateral substitution.

One simplified way to express synthetic leverage capacity is:

$$L_t = \frac{E_t + H_t + B_t}{C_t^{req}} \tag{4}$$

where:

- E_t = equity capital committed
- H_t = hedge-based synthetic exposure
- B_t = borrow-based exposure
- C_t^{req} = required collateral under prevailing settlement / margin rules

As settlement finality becomes faster and ownership verification becomes stricter, C_t^{req} tends to rise relative to reusable balance-sheet elasticity, reducing feasible leverage.

A more intuitive form is:

$$L_t^{eff} \propto \frac{\text{rehypothecation capacity} \times \text{netting window}}{\text{verification rigidity}} \tag{5}$$

Confirmed: collateral rules, verification timing, and netting windows affect effective leverage conditions.

Interpretive: digital settlement tends to reduce balance-sheet flexibility tied to delayed reconciliation.

Inferred: digital settlement may reduce the numerator and increase the denominator in the stylized leverage relationship shown above.

12 Operational Dataset Design

To convert the thesis into an empirical system, this paper specifies a full SAI dataset schema.

12.1 Dataset Unit of Observation

The preferred unit of observation is a **security-date** or **event-date** panel.

- For market backtests: daily panel data.
- For filing analysis: event-level data merged into daily data.
- For cross-sectional screens: issuer-date snapshots.

12.2 Core Dataset Tables

The operational architecture is divided into five core tables:

1. `issuer_events`
2. `regulatory_events`
3. `market_state`
4. `borrow_state`
5. `sai_panel`

12.3 Table 1: issuer_events

Table 3: Schema for `issuer_events`

Variable	Definition
event_id	Unique identifier for issuer event

issuer_ticker	Security ticker or issuer identifier
issuer_name	Issuer name
event_date	Filing or action date
event_type	S-3, 8-K, Form D, warrant issuance, prospectus supplement, etc.
filing_accession	SEC accession number where applicable
capital_structure_flag	Binary marker for delivery-sensitive corporate action
tokenization_flag	Binary marker for digital / tokenized ownership relevance
warrant_flag	Binary marker for warrant-related action
convertible_flag	Binary marker for convertible-related action
exercise_proceeds_est	Estimated potential proceeds from exercise or conversion
deliverable_change_flag	Binary marker for OCC / deliverable consequence
issuer_score	Researcher-assigned issuer significance score
notes	Free text / classification notes

12.4 Table 2: regulatory_events

Table 4: Schema for `regulatory_events`

Variable	Definition
reg_event_id	Unique identifier for regulatory event
agency	SEC, NSCC, FICC, OCC, DTCC, etc.
rule_id	Rule filing identifier
event_date	Filing / effective date

event_category	modernization, collateral, digital asset compatibility, settlement cycle, etc.
immediate_effect_flag	Binary indicator for immediate-effect designation
collateral_impact_score	Researcher score for collateral implications
settlement_compression_score	Researcher score for settlement-speed implications
digital_readiness_score	Researcher score for digital settlement relevance
notes	Free text

12.5 Table 3: market_state

Table 5: Schema for `market_state`

Variable	Definition
date	Trading date
ticker	Security ticker
close	Daily close price
volume	Daily volume
realized_vol_20d	20-day realized volatility
iv_atm_30d	At-the-money 30-day implied volatility
option_oi_calls	Aggregate call open interest
option_oi_puts	Aggregate put open interest
dealer_gamma_proxy	Model-based or estimated dealer gamma proxy
ftd_proxy	Fail-to-deliver proxy or actual value where available
short_interest_proxy	Short interest estimate or reported value
return_1d	Daily return
return_5d	5-day forward return
return_20d	20-day forward return

12.6 Table 4: borrow_state

Table 6: Schema for `borrow_state`

Variable	Definition
date	Observation date
ticker	Security ticker
shares_available	Reported or sampled borrow availability
borrow_fee	Annualized borrow fee
utilization_proxy	Utilization or tightness estimate
recall_flag	Binary indicator of borrow recall event
days_to_cover_proxy	Short-interest-to-volume proxy
locate_tightening_flag	Binary indicator for sharp availability compression

12.7 Table 5: sai_panel

Table 7: Schema for `sai_panel`

Variable	Definition
date	Observation date
ticker	Security ticker
R_score	Regulatory modernization component
D_score	Digital trading / ATS component
C_score	Clearing compression component
T_score	Tokenized issuance component
I_score	Issuer sensitivity component
SAI_raw	Unscaled weighted sum
SAI_z	Cross-sectional or time-series standardized SAI

squeeze_risk_score	Downstream model output
regime_label	Low / medium / high settlement-compression regime

Confirmed: the dataset design and variable definitions below are model architecture, not observed data.

Interpretive: the schema is designed to make the thesis testable.

Inferred: if populated consistently, the framework may detect whether settlement-compression regimes have explanatory value.

13 Scoring Methodology

Each component should be normalized to a common scale, such as $[0, 1]$ or z-score standardized.

13.1 Example Weighted Form

$$SAI_{i,t} = 0.25R_{i,t} + 0.20D_{i,t} + 0.20C_{i,t} + 0.15T_{i,t} + 0.20I_{i,t} \tag{6}$$

These weights are placeholders. Researchers can estimate optimal weights using one of three approaches:

a. equal or theory-driven weighting,

b. principal component analysis,

c. predictive optimization against downstream outcomes such as volatility spikes, borrow squeezes, or abnormal returns.

13.2 Illustrative Scoring Rubric

Table 8: Illustrative SAI Component Scoring Rubric

Component	Example Inputs
R	Rule modernization, accelerated settlement proposals, digital compatibility language, immediate-effect status
D	Tokenized ATS developments, blockchain-based transfer framework, digital custody relevance
C	Collateral compression, faster margin recognition, netting window reduction, operational settlement change
T	Warrant issuance, tokenized share references, digitally mapped ownership instruments
I	Low float, tight borrow, high short interest, complex deliverables, concentrated holder base

14 Event Study Framework

The SAI can be tested in an event-study setting. Let event windows be defined around material filing or rule dates. Then estimate:

$$AR_{i,t} = R_{i,t}^{actual} - R_{i,t}^{expected} \tag{7}$$

and cumulative abnormal return:

$$CAR_{i,[\tau_1,\tau_2]} = \sum_{t=\tau_1}^{\tau_2} AR_{i,t} \tag{8}$$

One can then test whether higher-SAI events are associated with:

- abnormal borrow tightening,

- increased implied volatility,
- changes in dealer gamma regime,
- elevated return dispersion,
- increased settlement-friction proxies.

Confirmed: event studies can be used to test market reactions around dated public disclosures.

Interpretive: the SAI offers a way to classify such disclosures into structural regimes.

Inferred: if high-SAI events systematically correlate with borrow or volatility stress, settlement modernization may function as a meaningful state variable.

15 Monte Carlo Squeeze-Risk Model

To operationalize market-structure vulnerability, the paper proposes a Monte Carlo model where price, volatility, and borrow evolve jointly.

15.1 Price Process

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t + \kappa Q_t dt \tag{9}$$

where:

- S_t = stock price
- μ = drift
- σ_t = time-varying volatility
- dW_t = Brownian shock
- Q_t = squeeze-induced order-imbalance term
- κ = price impact coefficient

15.2 Borrow Dynamics

$$B_t = B_{t-1} \exp(-\lambda_t + \eta_t) \tag{10}$$

where:

- B_t = shares available to borrow
- λ_t = deterministic tightening / recall factor
- η_t = stochastic shock

15.3 Dealer Feedback Term

A simple feedback approximation is:

$$Q_t = \phi_1 \max(0, -\Gamma_t)\Delta S_t + \phi_2 \mathbf{1}(B_t < B^{crit}) + \phi_3 \mathbf{1}(SAI_t > c) \tag{11}$$

15.4 Squeeze Trigger

Define a squeeze event as:

$$\mathcal{S}_t = \mathbf{1}\left(\frac{ShortDemand_t}{B_t + \omega V_t} > \theta\right) \tag{12}$$

where:

- $ShortDemand_t$ = estimated demand to maintain or close short exposure
- B_t = borrow supply
- V_t = secondary liquidity term
- ω = conversion coefficient for tradable liquidity
- θ = critical stress threshold

Confirmed: the model is a stylized simulation framework.

Interpretive: it combines price dynamics, borrow constraints, and dealer hedging in a way consistent with known microstructure stress channels.

Inferred: simulation outputs should be treated as scenario estimates rather than proof of expected market outcomes.

16 Backtesting Design

To test whether the SAI has empirical content, the framework supports three levels of backtesting.

16.1 Level 1: Structural Validation

Verify that known modernization events produce visible jumps in component scores.

16.2 Level 2: Market Response Validation

Test whether high-SAI periods are associated with:

- rising implied volatility,
- rising borrow fees,
- declining share availability,
- higher odds of outsized returns,
- changes in option-skew structure.

16.3 Level 3: Regime Model Validation

Classify days into low, medium, and high settlement-compression regimes and compare realized outcomes:

$$P(\text{squeeze}_{t+h} = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 SAI_t + \beta_2 BorrowTight_t + \beta_3 NegGamma_t + \beta_4 ShortProxy_t) \tag{13}$$

17 Risk Factors and Limits

This framework has important limitations.

1. Public data do not fully reveal synthetic exposure.
2. Borrow data are sampled and may be incomplete.
3. Tokenization language in public documents may signal exploration rather than implementation.
4. Rule filings may precede actual operational adoption by a wide margin.
5. Some scoring inputs require researcher judgment and should therefore be documented carefully.

Accordingly, the SAI should be interpreted as a *structural readiness index*, not direct proof of hidden positioning.

18 Investor and Policy Implications

If digitally verified settlement expands materially, several implications follow.

Confirmed: faster settlement and clearer ownership mapping can alter post-trade operations and compliance obligations.

Interpretive: securities lending and short maintenance may become more transparent, while delivery-sensitive capital structures may become more consequential for market participants.

Inferred: equities with concentrated ownership, low float, or complex deliverables may experience more convex market reactions because settlement friction can no longer be deferred as easily.

From a policy perspective, faster settlement is not automatically stabilizing. It reduces opacity, but may also compress adaptation time and increase intraday liquidity demands. The key distinction is that risk becomes more visible, not necessarily smaller.

19 Conclusion

The events reviewed in this paper are consistent with a structural transition in market infrastructure.

Confirmed: issuer actions, clearing modernization efforts, and digital trading infrastructure developments all occurred within the studied period.

Interpretive: these developments increasingly point in the same direction: reduced settlement opacity and higher ownership verifiability.

Inferred: if the trend continues and becomes operationally integrated, it could alter the economics of leverage, short maintenance, and dealer hedging in meaningful ways.

The clearest academically defensible conclusion is not that all actors were deliberately synchronized, but that multiple public developments appear directionally aligned with a more compressed and auditable settlement regime. The SAI framework proposed here provides a way to quantify that transition and test whether settlement modernization is becoming a meaningful state variable in U.S. equity market behavior.

20 Figure Placeholders

- Figure 1. Timeline of issuer and regulatory modernization events.
- Figure 2. SAI component stack and scoring flowchart.
- Figure 3. Borrow availability versus borrow fee under settlement-compression regimes.
- Figure 4. Monte Carlo squeeze probability distribution by SAI regime.
- Figure 5. Event-study cumulative abnormal returns around high-SAI events.
- Figure 6. Conceptual map of capital structure, settlement compression, and dealer hedging feedback.

A Appendix A: Operational Dataset Template

The following CSV header set provides a workable minimal schema for implementation.

```
date,ticker,issuer_name,event_id,event_type,filing_accession,agency,rule_id,
    event_category,capital_structure_flag,tokenization_flag,warrant_flag,
    convertible_flag,deliverable_change_flag,exercise_proceeds_est,R_score,D_score
    ,C_score,T_score,I_score,SAI_raw,SAI_z,close,volume,realized_vol_20d,
    iv_atm_30d,option_oi_calls,option_oi_puts,dealer_gamma_proxy,ftd_proxy,
    short_interest_proxy,shares_available,borrow_fee,utilization_proxy,recall_flag
    ,days_to_cover_proxy,locate_tightening_flag,return_1d,return_5d,return_20d,
    squeeze_risk_score,regime_label,notes
```

B Appendix B: Python Reference Implementation

```
import numpy as np
import pandas as pd
```

```
class SAIModel:
    def __init__(
        self,
        w_R=0.25,
        w_D=0.20,
        w_C=0.20,
        w_T=0.15,
        w_I=0.20,
    ):
        self.weights = {
            "R_score": w_R,
            "D_score": w_D,
            "C_score": w_C,
            "T_score": w_T,
            "I_score": w_I,
        }

    def compute_sai(self, df: pd.DataFrame) -> pd.DataFrame:
        required = list(self.weights.keys())
        missing = [c for c in required if c not in df.columns]
        if missing:
            raise ValueError(f"Missing required columns: {missing}")

        out = df.copy()
        out["SAI_raw"] = 0.0
        for col, wt in self.weights.items():
            out["SAI_raw"] += wt * out[col]
```

```
    std = out["SAI_raw"].std(ddof=0)
    mean = out["SAI_raw"].mean()
    out["SAI_z"] = 0.0 if std == 0 else (out["SAI_raw"] - mean) / std
    out["regime_label"] = out["SAI_z"].apply(self._classify_regime)
    return out

@staticmethod
def _classify_regime(sai_z: float, low_cut=-0.5, high_cut=0.5) -> str:
    if sai_z < low_cut:
        return "low"
    if sai_z > high_cut:
        return "high"
    return "medium"

@staticmethod
def compute_squeeze_risk(df: pd.DataFrame) -> pd.DataFrame:
    out = df.copy()

    def row_score(row):
        neg_gamma = max(0.0, -float(row.get("dealer_gamma_proxy", 0.0)))
        shares_available = max(float(row.get("shares_available", 1.0)), 1.0)
        borrow_fee = max(float(row.get("borrow_fee", 0.0)), 0.0)
        short_interest = max(float(row.get("short_interest_proxy", 0.0)),
0.0)
        sai_z = max(float(row.get("SAI_z", 0.0)), 0.0)

        borrow_tightness = (1.0 / shares_available) * 100000 + borrow_fee
```

```
            return (
                0.30 * neg_gamma
                + 0.30 * borrow_tightness
                + 0.20 * short_interest
                + 0.20 * sai_z
            )

        out["squeeze_risk_score"] = out.apply(row_score, axis=1)
        return out



def simulate_squeeze_paths(
    S0=25.0,
    B0=1_000_000,
    mu=0.00,
    sigma=0.60,
    neg_gamma=1.0,
    sai_level=1.0,
    lambda_b=0.05,
    days=20,
    n_sims=10000,
    critical_borrow=250_000,
    theta=1.25,
    seed=42,
):
    rng = np.random.default_rng(seed)

    dt = 1 / 252
    terminal_prices = np.zeros(n_sims)
```

```
squeeze_flags = np.zeros(n_sims, dtype=int)

for i in range(n_sims):
    S = S0
    B = B0
    squeeze = 0

    for _ in range(days):
        z = rng.normal()
        eps_b = rng.normal(scale=0.10)

        B = max(1.0, B * np.exp(-lambda_b + eps_b))

        feedback = 0.0
        if B < critical_borrow:
            feedback += 0.03
        feedback += 0.01 * max(neg_gamma, 0.0)
        feedback += 0.01 * max(sai_level, 0.0)

        dS = (mu + feedback) * S * dt + sigma * S * np.sqrt(dt) * z
        S = max(0.01, S + dS)

        short_demand = 1.0 + 0.25 * max(neg_gamma, 0.0) + 0.25 * max(
sai_level, 0.0)
        liquidity_supply = B / B0
        liquidity_supply = max(liquidity_supply, 1e-6)

        stress_ratio = short_demand / liquidity_supply
        if stress_ratio > theta:
```

```
            squeeze = 1

    terminal_prices[i] = S
    squeeze_flags[i] = squeeze

return {
    "squeeze_probability": float(squeeze_flags.mean()),
    "mean_terminal_price": float(terminal_prices.mean()),
    "median_terminal_price": float(np.median(terminal_prices)),
    "terminal_prices": terminal_prices,
    "squeeze_flags": squeeze_flags,
}
```

C Appendix C: Minimum Research Workflow

1. Parse issuer and regulatory filings into event tables.
2. Classify each event by SAI component relevance.
3. Merge event tables into daily security-level market data.
4. Compute SAI scores and standardized regimes.
5. Estimate borrow-tightness and dealer-gamma proxies.
6. Run event studies and regime-conditioned return tests.
7. Run Monte Carlo squeeze simulations under observed regime states.
8. Compare outputs against historical stress periods.

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